

The Puzzle of Deontic Presupposition

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Abstract

The principle of Inheritance says that *should* is closed under entailment: if ϕ entails ψ , then *should* ϕ entails *should* ψ . Inheritance is a fundamental principle of orthodox deontic logic and semantics. But Åqvist (1967) pointed out some apparent counterexamples to Inheritance: for example, *Max should know that he is breaking a rule* does not seem to entail *Max should be breaking a rule* even though knowledge is factive. I argue that Åqvist's observation is an instance of something more general: for many ϕ and ψ where ϕ entails and presupposes ψ , *should* ϕ does not entail *should* ψ . I argue that this phenomenon shows that the effective domain of a deontic modal is determined by the presuppositions of its prejacent along with certain presuppositions of the surrounding discourse. I formulate a semantics that incorporates this insight; the resulting theory accommodates the counterexamples to Inheritance while still preserving the truth of many instances of Inheritance.

Keywords: deontic modals, deontic paradoxes, Inheritance, presupposition

1 Introduction

I should wake up early tomorrow. I will wake up early tomorrow only if I go to bed now. So I should go to bed now. That reasoning seems impeccable. What principle underlies it? Perhaps this:

Instrumental Reasoning

should ϕ and (ϕ only if ψ) together entail *should* ψ .

Now consider the special case in which ϕ entails ψ and so the conditional (ϕ only if ψ) is itself a logical truth. In that case, Instrumental Reasoning implies that *should* ϕ entails *should* ψ . This yields the claim that *should* is closed under entailment, a principle variously called *Upward Monotonicity* or *Inheritance*:

Inheritance

If ϕ entails ψ , then *should* ϕ entails *should* ψ .

Inheritance has other sources of support as well. Certain instances of it seem obviously true: If I should wake up early and exercise, it would seem to follow that I should wake up early. From a more theoretical perspective, Inheritance is a fundamental principle of deontic logic and follows from standard approaches in natural language semantics: if all the deontically ideal worlds are ϕ -worlds and ϕ entails ψ , then all the deontically ideal worlds are ψ -worlds.

But here is an argument against Inheritance, due in its essentials to Åqvist (1967). Consider this case:

Known Violation

Max is breaking a school rule by eating in the hallway. He should not be doing that—he should not be breaking a school rule. But Max does not realize that he is breaking a rule: he does not know that he is not allowed to eat in the hallway. Still, Max *should know* the rules, and so he should know that he is breaking one of them.

In this scenario, the following sentence seems true:

- (1) Max should know that he is breaking a rule.

But since knowledge is factive, *Max knows that he is breaking a rule* entails *Max is breaking a rule*. So if Inheritance is true, (1) entails:

- (2) Max should be breaking a rule.

But (2) looks false: it is simply not true that Max should be breaking a rule. (Indeed, it seems that Max should *not* be breaking a rule.) If knowledge is factive and (1) is true while (2) is false, then Inheritance fails.

So there are considerations in favor of Inheritance: it follows from Instrumental Reasoning, it is supported by particular cases, it follows from standard approaches in logic and semantics. And yet Known Violation seems to suggest that Inheritance is false. This is the *Puzzle of Known Violation*.¹

Others have objected to Inheritance on other grounds (Jackson and Pargetter, 1986; Cariani, 2013), but I argue that the Puzzle of Known Violation presents a more formidable challenge to Inheritance than other prominent challenges to that principle (Section 2). When other apparent counterexamples are examined in more detail, they appear not to be counterexamples after all; rather they seem to result from the context-sensitivity of *should* (von Fintel, 2012; Blumberg and Hawthorne, 2023). But contextualist strategies that insulate Inheritance from other challenges do not extend to counterexamples like Known Violation; in cases like Known Violation, all the relevant data seem to tell against Inheritance.

The Puzzle of Known Violation turns out to be an instance of a more general puzzle, one we can call the *Puzzle of Deontic Presupposition* (Section 3). For many sentences ϕ and ψ such that ϕ

¹Åqvist (1967) calls this the *Epistemic Obligation Paradox*. Åqvist was less concerned about Inheritance in particular and more concerned about how to revise Standard Deontic Logic in light of cases like Known Violation.

entails and presupposes ψ , *should* ϕ does not seem to entail *should* ψ . Known Violation is a special case of this, since x *knows* ϕ entails and presupposes ϕ . Some other ‘deontic paradoxes’—such as the Good Samaritan Paradox (Prior, 1958)—also appear to be instances of the Puzzle of Deontic Presupposition. The interaction between deontic modals and presuppositions also poses a problem for another well-known principle of deontic reasoning, namely the principle of *Agglomeration* which says that *should* ϕ and *should* ψ together entail *should* (ϕ and ψ).

What do these puzzles teach us about the meaning of deontic modals? I first examine the prospects of handling the puzzle via conservative modifications of the standard semantics for deontic modals (Section 4). While these conservative theories fail, their failure is instructive and reveals an important pattern in the data, a pattern that points the way to a better theory (Section 5). The pattern in question is that, in an inference of the form *should* $\phi \Rightarrow$ *should* ψ , any presupposition of ϕ constrains the effective domain of an occurrence of *should* ψ unless that presupposition is logically related to ψ . I use this pattern to construct a more adequate theory, one that predicts that Inheritance fails in cases like Known Violation, but that also predicts that many instances of Inheritance are true. This allows us to account both for apparent violations of Inheritance and for the phenomena motivating Inheritance with which we began.

2 Challenges to Inheritance

Arguments against Inheritance are not new. Consider, for example, the case of Professor Procrastinate (Jackson and Pargetter, 1986):²

Professor Procrastinate

Professor Procrastinate is invited to review a book. He is the only fully qualified expert on the relevant topic and so the best possible outcome to this scenario is that Procrastinate accepts the invitation and writes the review. But Procrastinate procrastinates and he has a terrible time completing projects he undertakes. In particular, if Procrastinate accepts the invitation to review the book, it is extremely unlikely that he will complete the review. This is the worst possible outcome. If, on the other hand, Procrastinate declines, someone else will write the review. The alternative reviewer is less qualified, but will do a serviceable job.

The best outcome is that Procrastinate accepts and writes; the second best is that he declines; the worst outcome is that he accepts and fails to write. In light of these facts, Jackson and Pargetter argue that (3) is true while (4) is false:

- (3) Professor Procrastinate should accept the invitation and write the review.
- (4) Professor Procrastinate should accept the invitation.

²See also Cariani (2013).

Procrastinate should accept the invitation and write the review, for that is the optimal outcome. But should he accept the invitation? If you had to advise Procrastinate, I think the best advice would be to tell him that he should not accept—for if he accepts, he is unlikely to complete the review, which leads to the worst outcome in this scenario. So it seems that he should not accept the invitation, and that it is false that he should accept the invitation. But of course *Procrastinate accepts the invitation and writes the review* entails *Procrastinate accepts the invitation*. Thus, if (3) is true while (4) is false, it looks like the Procrastinate case is a counterexample to Inheritance.³

There is a well-known response to this argument open to advocates of Inheritance (von Fintel, 2012; Blumberg and Hawthorne, 2023). We sometimes encounter cases in which an utterance of ϕ looks true while an utterance of ψ looks false, even when ϕ really does entail ψ . This can happen when ϕ and ψ contain context-sensitive elements and the relevant parameters of the context are resolved differently across the two utterances. For example: an utterance of *Jumbo is a small elephant* might look true, while an utterance of *Jumbo is small* does not. But this does not necessarily show that the former does not entail the latter, for it might be that the context-sensitive adjective *small* is resolved differently across the two utterances. So we need to contend with the hypothesis that (3) is naturally evaluated in a certain type of context c , a type of context in which (3) is true (given the facts laid down in the case), while (4) is naturally evaluated in a different type of context c' , a type of context in which (4) is false (given those same facts). That could create the appearance that (3) does not entail (4) even when the entailment does in fact hold.

Here is how this story might go, given the standard semantics for deontic modals. Very roughly, the standard semantics—the semantics due to Angelika Kratzer—says that *should* ϕ is true at a context just in case ϕ is true in all the best of the relevant worlds, where what counts as ‘best’ and ‘relevant’ is determined by the context (Kratzer, 1977, 1981, 1991).⁴ Assume that ϕ entails ψ just in case: for all contexts c , if ϕ is true at c , then ψ is true at c . Then this semantics predicts that Inheritance holds in full generality. But it also allows the truth of *should* ϕ to vary with the context.

Suppose I utter (3), *Professor Procrastinate should accept the invitation and write the review*. Since I am evidently treating it as possible that he accepts and writes, it is natural to think that worlds in which he does both are relevant in this context. And it may be that all the best of the relevant worlds in this context are worlds in which Procrastinate accepts and writes—after all that is the best outcome. Now suppose we consider (4). When we evaluate (4), we may tend to focus more on what will happen if Procrastinate accepts, holding fixed what else we know about Procrastinate and his habits. And since it is unlikely that he both accepts and writes, we may end up ignoring worlds in which he accepts and writes. If no such worlds are relevant in *this* context, then it will not be true that all the best of the relevant worlds in this context are ones in which he accepts, since all the accept-worlds relevant in this context are accept-and-not-write-worlds. Thus,

³Ross’s Paradox (Ross, 1941) poses another challenge to Inheritance, but its source seems rather different. See Cariani (2013, 2021) and von Fintel (2012) for discussion of various challenges to Inheritance, including Ross’s Paradox.

⁴See also Lewis (1973).

(4) will be false in this context.

There are various questions one might ask about this particular contextualist story, and there may be ways to improve on it.⁵ But however the contextualist story is told, it is worth observing that there are compelling reasons to think that it is on the right track. For when we work to hold the context *fixed*, it is hard to endorse (3) while denying (4). To see this, note that when we conjoin (3) and the negation of (4), the result appears to be false (von Fintel, 2012; Blumberg and Hawthorne, 2023):

- (5) Professor Procrastinate should accept the invitation and write the review, but it's false that he should accept the invitation.

Imagine being Procrastinate and being told *You should accept the invitation and write the review, but it's false that you should accept*. It would be hard to know what to make of such advice. Similarly, consider the oddity of the following exchange:

- (6) a. A: Professor Procrastinate should accept the invitation and write the review.
b. B: So he should accept the invitation?
c. A: ? No, no, I didn't say that.

It's not easy to make sense of A's response here, which suggests that his initial utterance commits him to an affirmative answer to the question.

A natural response to these observations is to say that (3) really does entail (4). That would explain why (5) does not look to be true, and why A's final remark in (6) seems off. The intuition that (3) is true in the Procrastinate scenario while (4) is false could be explained in terms of context-shift in the manner suggested above. Thus, the totality of the data adduced here would arguably seem to support Inheritance rather than pose a problem for it.

An advocate of Inheritance impressed by the contextualist treatment of the Procrastinate case might wonder if it can be extended to cases like Known Violation. But here things look rather different, for all the relevant data in this case seem to tell against Inheritance.

As we noted earlier, in Known Violation, it looks as though (1) is true and (2) is false. Of course, given the possibility of a contextualist response, this observation is in principle compatible with Inheritance. But the contextualist account does not here enjoy the same support that it had in the Procrastinate case. Note, for example, that the following conjunction simply looks true:

- (7) Max should know that he is breaking a rule but he shouldn't be breaking a rule in the first place.⁶

⁵I prefer the slightly different contextualist account defended by Blumberg and Hawthorne (2023). But how exactly to best tell the contextualist story is somewhat ancillary to our main aim here, which is simply to establish that some sort of Inheritance-friendly contextualist account is motivated by the totality of the data surrounding the Procrastinate case.

⁶I assume that Max is not in a normative dilemma, so that if he should not be breaking a rule, it's false that he should be breaking a rule.

And *A*'s response in the following exchange is perfectly felicitous:

- (8) a. *A*: Max should know that he is breaking a rule.
 b. *B*: So he should be breaking a rule?
 c. *A*: No, no, of course not.

If there is something amiss with the above exchange, it is with *B*'s presuming that *A*'s initial remark in any way implies that Max should be breaking a rule.

Unlike in the Procrastinate case, then, all of the relevant data in the Known Violation case seem to tell against the hypothesis that (1) entails (2). Still, it is open to the advocate of Inheritance to maintain that, despite these observations, Inheritance really is valid and that (1) entails (2). But anyone defending this view would still need to explain the differences between the Procrastinate case and the Known Violation case that we have just highlighted. Perhaps they could explain the difference between (5) and (7) by saying that, in the former case, we find it difficult to interpret the two modals relative to different modal bases, whereas we find this easier to do in the latter case. But it is *prima facie* more natural, I think, to maintain that (1) does not entail (2). For that hypothesis explains all of our observations about the Known Violation case, while the hypothesis that (1) entails (2), by itself, explains little. Furthermore, as we shall see below, the pattern of apparent entailment and non-entailment we find in this vicinity admits of a precise characterization (Section 5.1), and systematic patterns of this sort are usually indicative of a semantic, rather than a pragmatic, phenomenon. In any case, the reader who would prefer to develop a pragmatic theory of these matters should nevertheless find the remainder of the essay useful as a guide to their own theorizing.

3 The Puzzle of Deontic Presupposition

Before moving on, I want to get a slightly clearer view of the empirical terrain. When philosophers say that *knows* is *factive*, they mean that *x knows that ϕ* entails ϕ . In linguistics, that property is sometimes called *veridicality*; to say that *knows* is *factive* in the linguist's sense is to say that *x knows that ϕ* presupposes ϕ . And *knows* does presuppose the truth of its complement, as the 'family-of-sentences' test reveals. Note that each of the following, if uttered in a normal context, would convey that it's raining, and would suggest that the speaker is taking it for granted that it's raining.⁷

- (9) a. Amy knows that it's raining.
 b. Amy doesn't know that it's raining.
 c. If Amy knows that it's raining, she will bring an umbrella.
 d. Amy might/must know that it's raining.

⁷Presuppositions do not always project out of these environments. The family-of-sentences test only checks to see if an inference projects out of various linguistic environments when the relevant sentences are considered in normal contexts. See Beaver et al. (2021).

- e. Does Amy know that it's raining?

It seems that other presupposition triggers also give rise to apparent failures of Inheritance. Consider other factive verbs, change-of-state verbs, and definite descriptions (I underline each presupposition trigger below):⁸

- (10) a. Boris should have remembered that Carl insulted Dido, but of course Carl shouldn't have insulted Dido in the first place.
 b. Elizabeth should regret that she was mean, and she shouldn't have been mean in the first place.
- (11) a. Gina should stop being mean, and she shouldn't be mean.
 b. Herman should have opened the window, and it shouldn't have been closed in the first place.
- (12) a. Poirot should catch the man who stole the jewels, but of course the jewels should not have been stolen.
 b. Ivan should apologize to the student he yelled at, but he should not have yelled at anyone.

In each of these cases the prejacent of the first conjunct entails and presupposes the negation of the prejacent of the second.⁹ Thus, these examples support the following generalization:

Presupposition Defeats Inheritance

For many ϕ and ψ such that ϕ entails and presupposes ψ , *should* ϕ does not entail *should* ψ .

Thus, the phenomenon exhibited in Known Violation is really an instance of a more general phenomenon, viz. the *Puzzle of Deontic Presupposition*.

We note that these examples also reveal that presuppositions project over *should*. For example, *Gina should stop being mean* itself presupposes that Gina has been being mean. This point will become important later in the essay.

It is worth mentioning that some of the other 'deontic paradoxes' also appear to be instances of the Puzzle of Deontic Presupposition. For example, standard formulations of the Good Samaritan Paradox (Prior, 1958) seem to have this same structure. McNamara and Van De Putte (2025) formulate the Good Samaritan Paradox by noting that Inheritance implies that (13) entails (14):

- (13) Jones should help Smith who has been robbed.
 (14) Smith should have been robbed.

⁸Again, I assume that in each of these cases, the subjects are not in normative dilemmas, so that if *should* (*not* ϕ) is true, so is *not* (*should* ϕ).

⁹Note that the order of the conjuncts does not seem to matter. For example, *Ivan should not have yelled at anyone, and he should apologize to the student he yelled at* is likewise felicitous. The reader may verify that the same is true of our other examples. In a sentence of the form *should* ϕ , ϕ is called the *prejacent*.

And yet (13) looks true while (14) looks false. And note that (15) looks consistent, which suggests that this is not merely a matter of context-shift:

- (15) Jones should help Smith who has been robbed, but Smith should not have been robbed.

This is an instance of the Puzzle of Deontic Presupposition on the assumption that non-restrictive relative clauses like *who has been robbed* trigger presuppositions.¹⁰

I do not claim that all deontic paradoxes are instances of the Puzzle of Deontic Presupposition. For example, Chisholm’s ‘Contrary to Duty’ Paradox (Chisholm, 1963) appears to concern how deontic modals interact with conditionals rather than how they interact with presuppositions. Ross’s Paradox (Ross, 1941) and the Puzzle of Free Choice Permission (Kamp, 1974) seem to concern disjunction and existential quantification rather than presupposition. How precisely to taxonomize the various deontic paradoxes is a task left for future inquiry.

Although the focus of this essay is on Inheritance, it is also worth noting that our puzzle cases can also be used to generate counterexamples to another well-known inference pattern, namely *Agglomeration*:¹¹

Agglomeration

should ϕ and *should* ψ entail *should* ϕ and ψ .

For the following look true in the Known Violation scenario:

- (16) It should be that Max knows that he is breaking a rule.
 (17) It should be that Max is not breaking a rule.

Given Agglomeration, these two sentences will entail:

- (18) It should be that: Max knows that he is breaking a rule and he is not breaking a rule.

But (18) looks false since it is impossible for Max to know that ϕ in a situation in which it is false that ϕ . The semantics developed below predicts these apparent failures of Agglomeration.

4 The standard theory and some variations

Earlier we said that, according to the standard semantics, *should* ϕ is true just in case ϕ is true at all of the best of the relevant worlds, where ‘best’ and ‘relevant’ are determined by context. Let’s make this more precise.

¹⁰The family-of-sentences test reveals that non-restrictive relative clauses project in the manner of presuppositions. But they may not, strictly speaking, count as *presuppositions* since they typically introduce new information to the discourse (Chierchia and McConnell-Ginet, 2000). It might be preferable to say that non-restrictive relative clauses conventionally mark some content at ‘not-at-issue’ and that not-at-issue content tends to project (Simons et al., 2010). So it may be that the correct generalization here is not in terms of presupposition, but in terms of not-at-issue content. This seems to me to be at least partly a terminological matter, and so I set it aside here.

¹¹For discussion of Agglomeration, see Jackson (1985), Cariani (2016), and Boylan (2023).

According to Kratzer (1981, 1991), modals are interpreted relative to two parameters, a *modal base* and an *ordering source*. A modal base is a function f that maps a world w to a non-empty set of relevant propositions $f(w)$. For Kratzer, the ‘relevant worlds’ at a context $(w, f, \dots)$ are the worlds at which all the propositions in $f(w)$ are true, i.e. $\bigcap f(w)$. We will simplify things here and take a modal base to be a function f that maps a world w to a non-empty set of worlds $f(w)$ —so we write $f(w)$ where Kratzer would write $\bigcap f(w)$. An ordering source for Kratzer is also a function g that maps a world w to a set $g(w)$ of propositions, and this set $g(w)$ is then used to define a preorder on the space of worlds. We will again simplify things here and use the term *ordering source* for a total preorder $\succeq$ on the space of worlds, where $w \succeq v$ means that w is at least as good as v (in the relevant respects determined by the context).¹² We represent contexts as triples $(w, f, \succeq)$ consisting of a world w , a modal base f , and an ordering source $\succeq$.

To state a version of Kratzer’s theory, it will help to adopt some notation. So let us write “ $w \succ v$ ” to mean that $w \succeq v$ and $v \not\succeq w$; thus $w \succ v$ iff w is strictly better than v (in relevant respects). And let us assume that $\succ$ is well-founded, which means that there are no infinite chains $w_1 \prec w_2 \prec w_3 \prec \dots$ of worlds such that each is strictly better than its predecessor (this is a ‘limit assumption’). Thus, for any non-empty set S of worlds, we can be sure that there is a non-empty set of best worlds in S , given any ordering source; we define such sets as follows:

$$\text{TOP}_{\succeq}(S) := \{w \in S : \neg \exists w' \in S : w' \succ w\}.$$

With all that in place, we can state a version of Kratzer’s theory as follows:

$$\llbracket \textit{should } \phi \rrbracket^{w, f, \succeq} = 1 \text{ iff for all } w' \in \text{TOP}_{\succeq}(f(w)), \llbracket \phi \rrbracket^{w', f, \succeq} = 1.$$

4.1 A non-entailment requirement?

As we noted earlier, this theory validates Inheritance, and so predicts, for example, that (1) entails (2). But von Fintel and Heim (2023, 64-65) suggest a solution to the Good Samaritan Paradox within this framework, and their approach would seem to generalize to the other cases we’ve been discussing.¹³ Recall the problem posed by the Good Samaritan Paradox: the standard theory predicts that (13) entails (14) and yet it seems that the former might be true while the latter is false:

(13) Jones should help Smith who has been robbed.

(14) Smith should have been robbed.

von Fintel and Heim suggest that the standard account can avoid this unwelcome prediction by saying that *should* ϕ is semantically defective if ϕ is true throughout the worlds in the modal base. They note that this could be encoded as a presupposition or as some other ingredient of meaning.

¹²A preorder is a reflexive and transitive relation; a total preorder is a preorder that is connected, i.e. for all w, v , either $w \succeq v$ or $v \succeq w$.

¹³See Portner (2009, 82-83) for a related approach.

For the sake of concreteness, let's suppose that this 'non-entailment requirement' is encoded as a presupposition, understood as a condition on definedness. Then the proposal would look like this:

$\llbracket \textit{should } \phi \rrbracket^{w,f,\succeq}$ is defined iff for some $w' \in f(w)$, $\llbracket \phi \rrbracket^{w',f,\succeq} \neq 1$.

Where defined, $\llbracket \textit{should } \phi \rrbracket^{w,f,\succeq} = 1$ iff for all $w' \in \text{TOP}_{\succeq}(f(w))$, $\llbracket \phi \rrbracket^{w',f,\succeq} = 1$.

How does this proposal solve the problem? Let's assume that ϕ entails ψ iff: for all contexts $(w, f, \succeq)$, if ϕ is true at $(w, f, \succeq)$, ψ is true at $(w, f, \succeq)$. Then this proposal predicts that (13) does not entail (14). For choose a context $(w, f, \succeq)$ in which (i) all worlds in $f(w)$ are worlds in which Smith is robbed, (ii) some worlds in $f(w)$ are worlds in which Jones does not help Smith, but (iii) the best worlds in $f(w)$ are worlds in which Jones helps Smith. By (ii) and (iii), it follows that (13) is defined and true at $(w, f, \succeq)$, but by (i) it follows that (14) is not defined (and so not true) at $(w, f, \succeq)$.

So this theory predicts that (13) does not entail (14); and, for related reasons, it predicts that (1) does not entail (2). But that is only one half of our puzzle: we want to predict both the instances of Inheritance that are false *and* the instances which are true. But this proposal does not satisfy this second *desideratum* because it makes it too difficult for an inference to be valid; it over-generates invalidities. For example, take the Procrastinate inference from (3) to (4):

(3) Professor Procrastinate should accept the invitation and write the review.

(4) Professor Procrastinate should accept the invitation.

This should come out valid (for reasons discussed earlier), but is not predicted to be valid by this theory. To see this, choose a context $(w, f, \succeq)$ such that (i) $f(w)$ entails that Procrastinate accepts the invitation but does not entail that he writes a review, and (ii) all the worlds in $\text{TOP}_{\succeq}(f(w))$ are worlds in which he accepts and writes. Condition (i) ensures that (3) is defined relative to $(w, f, \succeq)$, while condition (ii) ensures that, if defined, (3) is true relative to $(w, f, \succeq)$. But condition (i) implies that (4) is undefined at $(w, f, \succeq)$ because it implies that all the worlds in the modal base $f(w)$ are accept-worlds.¹⁴ So (4) is not true at $(w, f, \succeq)$, and so (3) is not predicted to entail (4). This is not what we want.

Another worry for this proposal is that it doesn't account for the fact that the following are consistent:

(15) Jones should help Smith who has been robbed, but Smith should not have been robbed.

(19) Jones should help Smith who has been robbed, but it's false that Smith should have been robbed.

Let's say that a sentence is consistent just in case there is some context at which it is true. Take (15), for example. Let $(w, f, \succeq)$ be any context. Suppose the left conjunct of (15) is true at $(w, f, \succeq)$.

¹⁴Although the term "modal base" refers for us to a function f from worlds to sets of worlds, we often abuse the term so that it refers to the value of that function at the world of the context, $f(w)$.

Then every world w' in $\text{TOP}_{\succeq}(f(w))$ is a world in which Jones helps Smith who has been robbed. But then every world w' in $\text{TOP}_{\succeq}(f(w))$ is a world in which Smith has been robbed. Note that, on our assumptions, $\text{TOP}_{\succeq}(f(w))$ is always non-empty. So some world w' in $\text{TOP}_{\succeq}(f(w))$ is a world in which Smith has been robbed, which means that the right conjunct of (15) is defined and false at $(w, f, \succeq)$. So (15) itself is false at $(w, f, \succeq)$. Since $(w, f, \succeq)$ was an arbitrary context, (15) is predicted to be inconsistent.

4.2 Presupposition-restricted modal bases

I don't see a simple fix for the above proposal. But there is another relatively simple modification of the standard theory that accounts for all of the observations discussed thus far. Unfortunately, even this theory is not quite right, but it is worth considering because it will take us in the direction of a better theory.

Consider these examples again:

- (1) Max should know that he is breaking a rule.
- (20) Gina should stop being mean.

Each of these has a certain 'contrary to duty' flavor in that they seem to say what someone should do on the assumption that certain (undesirable) facts obtain. Sentence (1) seems to say that, given that he is breaking a rule, Max should know that he is breaking a rule; (20) seems to say that, given that Gina is or was being mean, she should stop being mean.

Now suppose it is possible to consider (1) in a context whose modal base contains worlds in which Max is not breaking a rule. A natural thought is that when we consider (1) at such a context, we simply ignore such worlds. That is, we don't consider all the worlds in the relevant modal base and ask about what's happening in the best of those worlds; rather, we restrict our attention to the worlds in the modal base in which Max is breaking a rule and ask about what's happening in the best worlds in that restricted subset. When we utter (1) we are saying that the best worlds in the modal base in which Max is breaking a rule are all worlds in which he knows that he is breaking a rule; we are not saying that the best worlds in the modal base *simpliciter* are all worlds in which he knows that he is breaking a rule.

This is a plausible idea and it is straightforward to implement. Given a sentence ϕ , let $\Pi(\phi)$ be the set of propositions that ϕ presupposes. If ϕ has no presuppositions, then $\Pi(\phi)$ is simply the empty set; we assume in that case that $\bigcap \Pi(\phi) = \mathcal{W}$, the set of all possible worlds. Then we may offer the following modified semantics for *should*:

$$\llbracket \text{should } \phi \rrbracket^{w, f, \succeq} = 1 \text{ iff for all } w' \in \text{TOP}_{\succeq}(f(w) \cap \bigcap \Pi(\phi)), \llbracket \phi \rrbracket^{w', f, \succeq} = 1.$$

Let's say that the *effective domain* for *should* ϕ at a context $(w, f, \succeq)$ is the set that the ' $\text{TOP}_{\succeq}$ ' operation operates on. So on the standard theory, the effective domain for *should* ϕ is simply the modal base, $f(w)$. On the theory presently being entertained, the effective domain for *should* ϕ

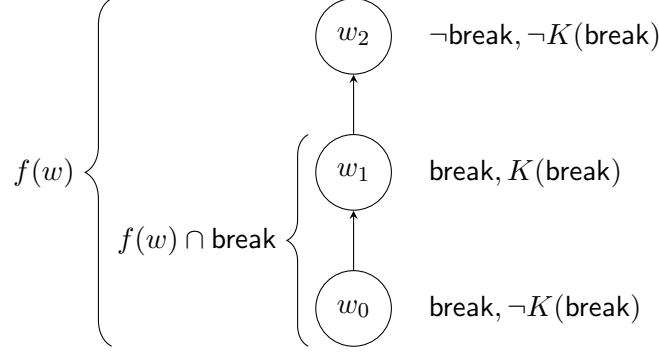


Figure 1: Breaking and knowing

is instead the modal base restricted to the set of worlds in which ϕ 's presuppositions are true, i.e. $f(w) \cap \bigcap \Pi(\phi)$.

Note an architectural difference between the standard theory and the present theory: on the standard theory the effective domain for *should* ϕ at a context is simply fixed by the context, and so it is the same for all choices of ϕ . On the present theory, in contrast, the effective domain for *should* ϕ at a context is partly a function of ϕ , and so will not necessarily be the same for all ϕ . In particular, if ϕ and ψ have different presuppositions, then the corresponding effective domains may well be different. It is this feature of the theory that allows it to predict failures of Inheritance, as we shall presently demonstrate.

This theory predicts our main two target facts: (i) that (1) does not entail (2) and (ii) that (3) entails (4).

To establish point (i), we need to find a context in which (1) is true while (2) is false. Let $(w, f, \succeq)$ be a context where $f(w)$ can be depicted by Figure 1. So the modal base $f(w)$ contains three worlds with each world strictly better than the one below it: $w_2 \succ w_1 \succ w_0$.

Then (1) is true at $(w, f, \succeq)$. Because *Max knows that he is breaking a rule* presupposes that Max is breaking a rule, the effective domain for (1) is $f(w) \cap \text{break}$, the result of intersecting the modal base $f(w)$ with **break**, the set of worlds in which Max is breaking a rule. The best world in $f(w) \cap \text{break}$ (there is only one) is w_1 and this is a world in which *Max knows that he is breaking a rule* is true, and so (1) is true at $(w, f, \succeq)$.

But (2), *Max should be breaking a rule*, is false at $(w, f, \succeq)$. To see this, note that if a sentence ϕ has no presuppositions, $\Pi(\phi)$ is empty, and so $\bigcap \Pi(\phi) = \mathcal{W}$. But then we have:

$$f(w) \cap \bigcap \Pi(\phi) = f(w) \cap \mathcal{W} = f(w).$$

Thus, since *Max is breaking a rule* does not have any relevant presuppositions, the effective domain for (2) is simply the modal base $f(w)$. And the best world in the modal base (there is only one) is w_2 and this is a world in which Max is not breaking a rule. So (2) comes out false at $(w, f, \succeq)$, and so (1) does not entail (2) according to this proposal.

This proposal predicts the invalidity of the inference from (1) to (2) by allowing the effective

domain of the premise to come apart from the effective domain of the conclusion. But this proposal does not always permit the effective domains of two occurrences of *should* to come apart in this way. In particular, if ϕ and ψ have the same presuppositions, then *should* ϕ and *should* ψ will have the same effective domain relative to any context $(w, f, \succeq)$, for if $\Pi(\phi) = \Pi(\psi)$, then $f(w) \cap \Pi(\phi) = f(w) \cap \Pi(\psi)$. Thus, if ϕ entails ψ in such a case, *should* ϕ will entail *should* ψ . The theory thus predicts that many instances of Inheritance are true. For example: the theory predicts that the Procrastinate inference, the inference from (3) to (4) is valid. For in the Procrastinate inference, neither prejacent has any relevant presuppositions, and so the two prejacentes have the same presuppositions.

The present account also predicts the consistency of (15) and (19), as the reader may verify. What allows it to do this is that, even at a fixed context $(w, f, \succeq)$, the effective domains of the two conjuncts of a conjunction *should* ϕ and *should* ψ may differ, since the effective domain for each conjunct is sensitive to its prejacent. Thus, the present theory handles all of the observations we've discussed so far.

But while this account does look like an improvement, it still faces a problem: it still over-generates invalidities. The problem is that whenever the premise and conclusion of an inference of the form *should* $\phi \Rightarrow$ *should* ψ differ in terms of what their prejacentes presuppose, the present account threatens to invalidate the inference. But sometimes such inferences are valid. Consider, for example, the inference from (21) to (22):

(21) Max should know that he is breaking a rule and apologize to the headmaster.

(22) Max should apologize to the headmaster.

This inference appears to be valid (imagine affirming the premise while denying the conclusion). But it is not predicted to be valid on the present account.

To see this, note the prejacent of (21), *Max knows that he is breaking a rule and apologizes to the headmaster* presupposes that Max is breaking a rule. (This claim seems intuitively correct and standard theories of presupposition projection predict that presuppositions project out of left conjuncts.) The prejacent of (22), *Max apologizes to the headmaster*, on the other hand, does not seem to have any relevant presuppositions. Thus, the effective domain of (21) at a context $(w, f, \succeq)$ will be the modal base restricted by the proposition that Max is breaking a rule, $f(w) \cap \text{break}$, while the effective domain of (22) at $(w, f, \succeq)$ will simply be the modal base $f(w)$. So while it could be that the best worlds in $f(w) \cap \text{break}$ are all worlds in which Max knows that he is breaking a rule and apologizes to the headmaster, it may be that not all the best worlds in $f(w)$ are worlds in which Max apologizes to the headmaster. Perhaps the best worlds in the modal base are all ones in which Max does nothing for which he needs to apologize. But given that he is breaking a rule, he should know that he is and also apologize to the headmaster. Structurally, the sort of context that invalidates this inference is similar to the one depicted in Figure 1; we depict such a context in Figure 2 (again, each world is strictly better than the one below it). The reader can verify that, on the semantics given above, the depicted context $(w, f, \succeq)$ makes (21) true and (22) false.

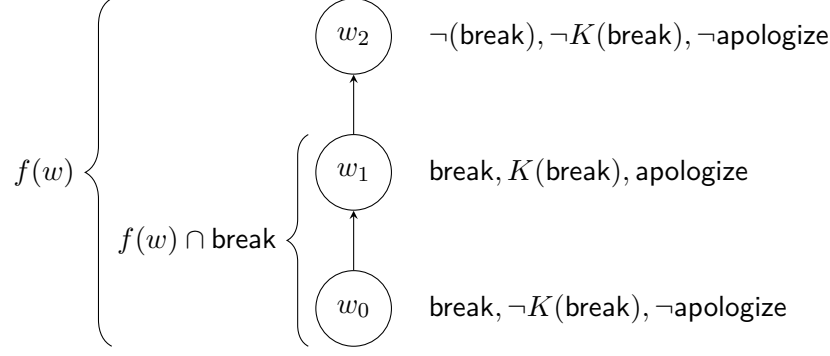


Figure 2: Breaking, knowing, apologizing

5 The meaning of *should*

5.1 A pattern

Where did this last proposal go wrong? Suppose we are considering an inference of the form *should* $\phi \Rightarrow$ *should* ψ where ϕ and ψ differ in what they presuppose. It seems that in some such cases, we want the effective domain for the premise to come apart from the effective domain of the conclusion, for this is what allows us to invalidate the inference from (1) to (2). But in other such cases, it seems we want the effective domain for the premise to coincide with the effective domain of the conclusion—that, at least, is one conclusion to draw from the apparent validity of the inference from (21) to (22). The problem with the present theory is that it allows the effective domain of the premise to come apart from the effective domain of the conclusion whenever premise and conclusion differ in terms of what they presuppose. It looks like we need a more fine-grained story about when the effective domains of two modal sentences can come apart and when they must coincide.

Suppose we continue to assume that the effective domain of *should* ϕ is constrained by ϕ 's presuppositions. But now we add: the effective domain of an occurrence of *should* ϕ may be constrained not just by ϕ 's presuppositions but also by the presuppositions of the *surrounding discourse*. Think of a discourse as a set of sentences; to say that a discourse Γ presupposes ψ is to say that some sentence in Γ presupposes ψ . In an inference, we may assume that 'the surrounding discourse' for the conclusion is the set of premises. And we may suppose that if *should* ϕ occurs with surrounding discourse Γ , and Γ presupposes ψ , then, as a default, ψ will constrain the effective domain of *should* ϕ , i.e. all the worlds in the effective domain for *should* ϕ will be ψ -worlds.

Now suppose this default is in place in the inference from (21) to (22), repeated here:

- (21) Max should know that he is breaking a rule and apologize to the headmaster.
- (22) Max should apologize to the headmaster.

Then since (21) presupposes that Max is breaking a rule, that presupposition will constrain both the effective domain of the premise *and* the effective domain of the conclusion. That is, in a context

$(w, f, \succeq)$, the effective domain of both premise and conclusion will be the same, namely $f(w) \cap \text{break}$. And if (21) and (22) have the same effective domain at any context, the inference from the former to the latter will be valid, since any world in which Max is breaking a rule and apologizes to the headmaster is a world in which he apologizes to the headmaster.

But of course we don't want the presuppositions of the surrounding discourse to *always* constrain the effective domain of a sentence of the form *should* ϕ —that default should merely be a default. For we want to preserve the above explanation of why the inference from (1) to (2) fails, the explanation depicted by Figure 1. According to that explanation, the effective domain of the premise is constrained by the presupposition *Max is breaking a rule*, but the effective domain of the conclusion is *not* constrained by that presupposition, even though it is a presupposition of the surrounding discourse for the conclusion. So we need to allow that sometimes the effective domain of an occurrence of *should* ϕ is not constrained by a particular presupposition of the surrounding discourse. The question then becomes: when does a presupposition of the surrounding discourse constrain the effective domain of *should* ϕ and when does it not?

To arrive at an answer to this question, it will help to examine some inferences more closely, for when we do this, a pattern begins to emerge. To see this pattern, it will help to represent those inferences using some additional notation. We will write ϕ_ψ to indicate a situation in which ψ is an elementary presupposition of ϕ (Beaver and Krahmer, 2001). So, for example, if b translates *Max is breaking a rule*, then we'll use Kb_b to translate *Max knows that he is breaking a rule*, the translation indicating that this sentence presupposes that Max is breaking a rule. And for any sentence ϕ , we use $[\phi]$ to denote the proposition (set of worlds) expressed by ϕ .

Start with the valid inference from (21) to (22), translated into this notation:

- (21) Max should know that he is breaking a rule and apologize to the headmaster.
should ($Kb_b \wedge a$)
- (22) Max should apologize to the headmaster.
should a

Now the premise presupposes $[b]$, the proposition that Max is breaking a rule. This is because presuppositions project out of left conjuncts, and also project over deontic *should*. Now notice the following fact about how the prejacent of the conclusion, $a = \text{Max apologizes to the headmaster}$, relates to b : a and b are logically independent. By this, I mean that a entails neither b nor $\neg b$ and b entails neither a nor $\neg a$: settling the truth value of either sentence does not settle the truth value of the other. So in the inference (21) $\Rightarrow$ (22), the prejacent of the conclusion and the presupposition of the premise are logically independent.

Now compare this to an invalid inference, one we have yet to discuss:¹⁵

- (21) Max should know that he is breaking a rule and apologize to the headmaster.

¹⁵Thanks to [NAME OMITTED] for drawing my attention to inferences like this one. This sort of inference posed a problem for a theory I advocated in an earlier version of this paper.

should ($Kb_b \wedge a$)

(23) Max should be breaking a rule and apologize to the headmaster.

should ($b \wedge a$)

This inference appears to be just as invalid as the inference from (1) to (2)—one cannot rescue the Known Violation inference simply by adding a conjunct to the premise and the conclusion. Now, again, the premise of this inference presupposes $[b]$. But look at how the prejacent of the conclusion, $(b \wedge a)$, relates to b this time: $(b \wedge a)$ entails b . So, this time, the prejacent of the conclusion and the presupposition of the premise are *not* logically independent.

This seems like an important difference between the valid inference from (21) to (22) and the invalid inference from (21) to (23). In fact, this pattern holds across all the inferences we’ve discussed so far: in the invalid inferences, there is a ϕ such that the premise presupposes $[\phi]$ and the prejacent of the conclusion and ϕ are ‘logically dependent’, whereas in the valid inferences there is no such ϕ .

Consider the invalid Known Violation inference from (1) to (2), represented as *should* $Kb_b \Rightarrow$ *should* b . In this inference, the premise presupposes $[b]$, and the prejacent of the conclusion, b , entails b , since any sentence entails itself. So there is a ϕ such that the premise presupposes $[\phi]$ and the prejacent of the conclusion and ϕ are logically dependent.

Finally, consider the valid Procrastinate inference from (3) to (4), represented as *should* $(i \wedge r) \Rightarrow$ *should* i . In this case, the premise has no relevant presuppositions. Thus, there is no ϕ such that $[\phi]$ is presupposed by the premise, and the prejacent of the conclusion and ϕ are logically dependent.

5.2 Background-restricted modal bases

Our pattern is this: in each of the invalid inferences we’ve examined, there is a ϕ such that the premise presupposes $[\phi]$ and ϕ and the prejacent of the conclusion are not logically independent; in the valid cases, there is no such ϕ . How can we formulate a theory of *should* that captures this pattern? The present section outlines our answer to this question; a more rigorous formulation of our semantics can be found in the appendix.

On the theory discussed in Section 4.2, the effective domain of a sentence of the form *should* ϕ in a context $(w, f, \succeq)$ was determined by intersecting the modal base $f(w)$ with the set of worlds in which ϕ ’s presuppositions were true, $\bigcap \Pi(\phi)$, i.e.:

$$f(w) \cap \bigcap \Pi(\phi).$$

On the present approach, we retain these two determinants of the effective domain. But we want to allow that the effective domain of an occurrence of *should* ϕ might be further constrained by some of the presuppositions of the surrounding discourse. Given the discussion above, we might hypothesize that the effective domain of an occurrence of *should* ϕ is constrained by those presuppositions of the surrounding discourse that are logically independent of ϕ . But it will simplify things if we

restrict this formulation slightly. Let's say that a sentence ϕ is *non-modal* just in case it contains no occurrences of *should*. If ϕ is a non-modal sentence, then we also say that the proposition $[\phi]$ expressed by ϕ is non-modal. Then our proposal is that the effective domain of an occurrence of *should* ϕ is constrained by the *non-modal* presuppositions $[\psi]$ of the surrounding discourse such that ϕ and ψ are logically independent. While this restriction might not be strictly necessary, it does simplify the presentation; note also that in all of our examples, the relevant presuppositions are non-modal.

We now speak of the effective domain of an occurrence of *should* ϕ relative to a context $(w, f, \succeq)$ and a *surrounding discourse* (set of sentences) Γ . As before, let $\Pi(\phi)$ be the set of propositions that ϕ presupposes. If Γ is a set of sentences, then we say that Γ presupposes a proposition iff some sentence in Γ presupposes that proposition, and we let $\Pi(\Gamma)$ be the set of propositions that Γ presupposes. And we let $\Pi_\phi(\Gamma)$ be the set of non-modal propositions $[\psi]$ in $\Pi(\Gamma)$ such that ψ and ϕ are logically independent, i.e.:

$$\Pi_\phi(\Gamma) = \{[\psi] : [\psi] \text{ is non-modal} \ \& \ [\psi] \in \Pi(\Gamma) \ \& \ \phi \text{ and } \psi \text{ are logically independent}\}.$$

Then the effective domain of *should* ϕ relative to a context $(w, f, \succeq)$ and a prior discourse Γ is:

$$f(w) \cap \bigcap \Pi(\phi) \cap \bigcap \Pi_\phi(\Gamma),$$

where:

- $f(w)$ is the modal base,
- $\bigcap \Pi(\phi)$ is the set of worlds that verify the presuppositions of the prejacent ϕ , and
- $\bigcap \Pi_\phi(\Gamma)$ is the set of worlds that verify the non-modal presuppositions of the surrounding discourse Γ that are logically independent of ϕ .

This yields the following semantics for *should*:

$$\llbracket \text{should } \phi \rrbracket^{w, f, \succeq, \Gamma} = 1 \text{ iff for all } w' \in \text{TOP}_{\succeq}(D(\phi)), \llbracket \phi \rrbracket^{w', f, \succeq, \Gamma} = 1,$$

$$\text{where } D(\phi) = f(w) \cap \bigcap \Pi(\phi) \cap \bigcap \Pi_\phi(\Gamma).$$

Note that our points of evaluation include both a context $(w, f, \succeq)$ and a parameter Γ that represents the surrounding discourse. The second component of this proposal is the account of entailment, which treats the premise set of an inference as the surrounding discourse for the conclusion:

Entailment

A set of sentences Γ *entails* ϕ iff: for any context $(w, f, \succeq)$, if every $\gamma \in \Gamma$ is such that $\llbracket \gamma \rrbracket^{w, f, \succeq, \Gamma} = 1$, then $\llbracket \phi \rrbracket^{w, f, \succeq, \Gamma} = 1$.

When the conclusion ϕ is of the form *should* ψ , this account implies that the presuppositions of the premise set that are logically independent of ψ will constrain the effective domain of *should* ψ .

5.3 Inheritance

Like the theory of Section 4.2, the present theory predicts that the Known Violation inference $(1) \Rightarrow (2)$ is invalid while the Procrastinate inference $(3) \Rightarrow (4)$ is valid. But it improves on that theory insofar as it predicts the validity of $(21) \Rightarrow (22)$.

The account of the inferences $(1) \Rightarrow (2)$ and $(3) \Rightarrow (4)$ that the present theory gives is quite similar to the account given by the theory of Section 4.2. In fact, the present theory in a sense collapses into that theory for inferences of the form *should* $\phi \Rightarrow$ *should* ψ in which there is no presupposition of the premise that is logically independent of ψ . And both $(1) \Rightarrow (2)$ and $(3) \Rightarrow (4)$ meet that condition. But let's unpack this more slowly.

Let's start by examining the inference from (1) to (2), represented as *should* $Kb_b \Rightarrow$ *should* b . Consider the context $(w, f, \succeq)$ illustrated in Figure 1 (from Section 4.2) and let Γ be the premise set, which in this case just means that $\Gamma = \{\textit{should } Kb_b\}$. The effective domain for the premise, $D(Kb_b)$, will be the result of intersecting the modal base $f(w)$ with the presuppositions of Kb_b and the non-modal presuppositions of the discourse $\Gamma = \{\textit{should } Kb_b\}$ that are logically independent of Kb_b :

$$D(Kb_b) = f(w) \cap \bigcap \Pi(Kb_b) \cap \bigcap \Pi_{Kb_b}(\{\textit{should } Kb_b\}).$$

The only presupposition of Kb_b is $[b]$, so $\bigcap \Pi(Kb_b) = [b]$. The presuppositions of *should* Kb_b are just those of Kb_b , i.e. $[b]$. But since Kb_b entails b , there are no non-modal presuppositions of $\{\textit{should } Kb_b\}$ that are logically independent of Kb_b , so $\Pi_{Kb_b}(\{\textit{should } Kb_b\}) = \emptyset$, which means $\bigcap \Pi_{Kb_b}(\{\textit{should } Kb_b\}) = \mathcal{W}$. So the effective domain of the premise, $D(Kb_b)$, is simply the result of intersecting the modal base $f(w)$ with $[b]$:

$$D(Kb_b) = f(w) \cap [b] \cap \mathcal{W} = f(w) \cap [b].$$

In Figure 1, $[b]$ is denoted by **break**, and $f(w) \cap \text{break} = \{w_0, w_1\}$. Since the best world in that set (there is only one) is w_1 and Kb_b is true at w_1 , it follows that *should* Kb_b is true at $(w, f, \succeq, \Gamma)$.

Turning now to the conclusion, note that the effective domain of *should* b , $D(b)$, will be the result of intersecting the modal base $f(w)$ with the presuppositions of b and the non-modal presuppositions of the discourse $\Gamma = \{\textit{should } Kb_b\}$ that are logically independent of b :

$$D(b) = f(w) \cap \bigcap \Pi(b) \cap \bigcap \Pi_b(\{\textit{should } Kb_b\}).$$

Thus, since b has no presuppositions, $\Pi(b)$ is empty, and so $\bigcap \Pi(b) = \mathcal{W}$. Again, the only non-modal presupposition of *should* Kb_b is $[b]$; but since b is not logically independent of b , $\Pi_b(\{\textit{should } Kb_b\})$ is likewise empty, and so $\bigcap \Pi_b(\{\textit{should } Kb_b\}) = \mathcal{W}$. Thus, the effective domain of the conclusion, $D(b)$, is just the modal base $f(w)$:

$$D(b) = f(w) \cap \mathcal{W} \cap \mathcal{W} = f(w).$$

In Figure 1, $f(w) = \{w_0, w_1, w_2\}$. Since w_2 is the unique best world in that set, and b is false at

w_2 , it follows that *should* b is false at $(w, f, \succeq, \Gamma)$. Thus, (1) does not entail (2) according to this approach.

When we consider *valid* inferences, it is helpful to note the following. Suppose we have an inference of the form *should* $\phi \Rightarrow$ *should* ψ , where ϕ entails ψ . Then in order to show that this inference is valid, it will suffice to show that the effective domain of the premise, $D(\phi)$, at any point $(w, f, \succeq, \Gamma)$ with $\Gamma = \{\textit{should } \phi\}$ is identical to the effective domain of the conclusion, $D(\psi)$, at that point. For in that case, if $(w, f, \succeq, \Gamma)$ is a point at which *should* ϕ is true, then every world in $\text{TOP}_{\succeq}(D(\phi))$ will be a world at which ϕ is true. But since ϕ entails ψ , every world in $\text{TOP}_{\succeq}(D(\phi))$ will be a world at which ψ is true. And since $D(\phi) = D(\psi)$, it will follow that every world in $\text{TOP}_{\succeq}(D(\psi))$ will be a world at which ψ is true. Thus, *should* ψ will also be true at $(w, f, \succeq, \Gamma)$.

The validity of the Procrastinate inference from (3) to (4), represented as *should* $(i \wedge r) \Rightarrow$ *should* i falls out of this observation. For let $(w, f, \succeq, \Gamma)$ be any point with $\Gamma = \{\textit{should } (i \wedge r)\}$. As the reader can verify, the fact that neither $(i \wedge r)$ nor i has any presuppositions means that both $D(i \wedge r)$ and $D(i)$ are identical to the modal base $f(w)$. Thus, since $D(i \wedge r) = D(i)$ at any point $(w, f, \succeq, \Gamma)$ with $\Gamma = \{\textit{should } (i \wedge r)\}$, the validity of *should* $(i \wedge r) \Rightarrow$ *should* i follows from the observation of the previous paragraph.

We can generalize this last result to obtain a restricted version of Inheritance. Because $(i \wedge r)$ and i have no presuppositions, they have the same presuppositions, i.e. $\Pi(i \wedge r) = \Pi(i)$. And it turns out that whenever this happens, the corresponding instance of Inheritance is true:

Restricted Inheritance

If ϕ entails ψ , and ϕ and ψ have precisely the same presuppositions (i.e. $\Pi(\phi) = \Pi(\psi)$), *should* ϕ entails *should* ψ .

(The proof of this can be found in the appendix.) This general result means that the present theory at least partly explains the appeal of Inheritance, insofar as it predicts that many instances of that principle are true.

Finally, we need to verify that the present theory really does improve on the theory of Section 4.2 by showing that it predicts the validity of the inference from (21) to (22), represented as *should* $(Kb_b \wedge a) \Rightarrow$ *should* a . So let $(w, f, \succeq, \Gamma)$ be any point with $\Gamma = \{\textit{should } (Kb_b \wedge a)\}$. In light of the above remarks, it will suffice to show that $D(Kb_b \wedge a) = D(a)$. Note that

$$D(Kb_b \wedge a) = f(w) \cap \bigcap \Pi(Kb_b \wedge a) \cap \bigcap \Pi_{Kb_b \wedge a}(\{\textit{should } (Kb_b \wedge a)\}).$$

Since the only presupposition of $Kb_b \wedge a$ is $[b]$, $\bigcap \Pi(Kb_b \wedge a) = [b]$. The only presupposition of *should* $(Kb_b \wedge a)$ is $[b]$, but since b is not logically independent of $Kb_b \wedge a$, $\Pi_{Kb_b \wedge a}(\{\textit{should } (Kb_b \wedge a)\}) = \emptyset$ and so $\bigcap \Pi_{Kb_b \wedge a}(\{\textit{should } (Kb_b \wedge a)\}) = \mathcal{W}$. Thus, we have:

$$D(Kb_b \wedge a) = f(w) \cap [b] \cap \mathcal{W} = f(w) \cap [b].$$

And note that

$$D(a) = f(w) \cap \bigcap \Pi(a) \cap \bigcap \Pi_a(\{should(Kb_b \wedge a)\}).$$

Since $\Pi(a)$ is empty, $\bigcap \Pi(a) = \mathcal{W}$. Since $[b]$ is the sole presupposition of $should(Kb_b \wedge a)$ and b and a are logically independent, $\bigcap \Pi_a(\{should(Kb_b \wedge a)\}) = [b]$. Thus,

$$D(a) = f(w) \cap \mathcal{W} \cap [b] = f(w) \cap [b].$$

Thus, $D(Kb_b \wedge a) = D(a)$, which is what we needed to show. Note how in this example, the presupposition of the premise $[b]$ constrains both the effective domain of the premise and the effective domain of the conclusion, and it does so precisely because b is logically independent of the prejacent of the conclusion, a .

5.4 Agglomeration

In Section 2, we noted that the following sentence seems true in the Known Violation scenario:

- (7) Max should know that he is breaking a rule but he shouldn't be breaking a rule in the first place.
should Kb_b ∧ should ¬b

On the standard semantics this sentence is inconsistent because there is no world at which Kb_b and $\neg b$ are both true. But on the present approach, this sentence comes out consistent, given the following natural definition of consistency:

Consistency

A set of sentences Γ is *consistent* iff there is a context $(w, f, \succeq)$ such that for all $\gamma \in \Gamma$, $\llbracket \gamma \rrbracket^{w, f, \succeq, \Gamma} = 1$. A sentence ϕ is consistent iff $\{\phi\}$ is consistent.

Note that here the discourse parameter is set to Γ , the very set of sentences being assessed for consistency.

The context $(w, f, \succeq)$ depicted in Figure 1 can be used to establish the consistency of (7), if we set the discourse parameter $\Gamma = \{should Kb_b \wedge should \neg b\}$. Note that (7)'s only presupposition is $[b]$. Since b is not logically independent of either prejacent, we have

$$\bigcap \Pi_{Kb_b}(\Gamma) = \bigcap \Pi_{\neg b}(\Gamma) = \mathcal{W}.$$

Since Kb_b presupposes only $[b]$, this means the effective domain for the first conjunct is $f(w) \cap [b]$. In Figure 1, $f(w) \cap [b] = \{w_0, w_1\}$. Since w_1 is the unique best world in that set, and since Kb_b is true at w_1 , $should Kb_b$ is true at $(w, f, \succeq, \Gamma)$. Since the prejacent of the second conjunct has no presuppositions, the effective domain of that conjunct is simply the modal base $f(w)$. Since the unique best world in the modal base is w_2 and $\neg b$ is true at w_2 , $should \neg b$ is also true at $(w, f, \succeq, \Gamma)$. Thus, we predict that (7) is consistent.

A corollary of this argument is that we predict that the following instance of Agglomeration is invalid:

- (16) It should be that Max knows that he is breaking a rule.
should Kb_b
- (17) It should be that Max is not breaking a rule.
should $\neg b$
- (18) It should be that: Max knows that he is breaking a rule and he is not breaking a rule.
should $(Kb_b \wedge \neg b)$

This prediction is welcome, since, as we noted in Section 3, sentences (16) and (17) look true in the Known Violation scenario, while (18) looks false. Again, take the Figure 1 context $(w, f, \succeq)$ and set $\Gamma' = \{\textit{should } Kb_b, \textit{should } \neg b\}$. Since the only presupposition of Γ' is $[b]$ and b is not logically independent of any prejacent in this inference, we have:

$$\bigcap \Pi_{Kb_b}(\Gamma') = \bigcap \Pi_{\neg b}(\Gamma') = \bigcap \Pi_{Kb_b \wedge \neg b}(\Gamma') = \mathcal{W}.$$

Thus, as above, the effective domain of *should* Kb_b is $f(w) \cap [b]$ and the effective domain of *should* $\neg b$ is $f(w)$, and so both of these sentences are true at $(w, f, \succeq, \Gamma')$ for the same reason their conjunction is true at $(w, f, \succeq, \Gamma)$ (see the previous paragraph). But the conclusion is false at $(w, f, \succeq, \Gamma')$. The conclusion only presupposes $[b]$, so the effective domain of the conclusion is $f(w) \cap [b] = \{w_0, w_1\}$. But $Kb_b \wedge \neg b$, being a contradiction, is not true at either of w_0 or w_1 . It follows that *should* $(Kb_b \wedge \neg b)$ is false at $(w, f, \succeq, \Gamma')$. Thus, this instance of Agglomeration fails, as desired.

6 Conclusion

We began with Åqvist's Puzzle of Known Violation, arguing that it poses a more serious threat to Inheritance than other well-known challenges to that principle, such as the Professor Procrastinate case. The totality of the Procrastinate data arguably supports a standard contextualist semantics for *should*, whereas the totality of the Known Violation data seems to tell against Inheritance. We then generalized Åqvist's puzzle, arguing that for many ϕ and ψ such that ϕ entails and presupposes ψ , *should* ϕ does not seem to entail *should* ψ . Finally, we used this general Puzzle of Deontic Presupposition to motivate a semantics on which the effective domain of a sentence of the form *should* ϕ is determined by three factors: the contextually determined modal base, the presuppositions of ϕ , and the presuppositions of the surrounding discourse that are logically independent of ϕ .

It is a general feature of the standard semantics that the effective domain of a deontic modal is partly determined by context. It is thus not too surprising that the surrounding discourse—the sentences used in a context—can influence the effective domain of a deontic modal, since the surrounding discourse is, in some sense, part of the context of utterance. But what is notable here

is how systematic this phenomenon is, as revealed by the pattern discussed in Section 5.1. Thus, whether or not the precise semantic account developed here is the right one, any future account that improves on it will need to take account of this pattern.

Appendix

Definition 1. We assume a vocabulary that includes atomic sentences (p , q , etc.) and other elements as indicated below. We define two languages, one that subsumes the other:

1. The *sentences of $\mathcal{L}_0$* are as follows:

- (i) Every atomic sentence is a sentence of $\mathcal{L}_0$.
- (ii) If ϕ and ψ are sentences of $\mathcal{L}_0$, then $\neg\phi$, $(\phi \wedge \psi)$, ϕ_ψ , and $K\phi$ are sentences of $\mathcal{L}_0$.
- (iii) Nothing else is a sentence of $\mathcal{L}_0$.

2. The *sentences of $\mathcal{L}$* are as follows:

- (i) Every sentence of $\mathcal{L}_0$ is a sentence of $\mathcal{L}$.
- (ii) If ϕ is a sentence of $\mathcal{L}_0$, then *should* ϕ is a sentence of $\mathcal{L}$.
- (iii) If ϕ and ψ are sentences of $\mathcal{L}$, then $\neg\phi$, $(\phi \wedge \psi)$, ϕ_ψ , and $K\phi$ are sentences of $\mathcal{L}$.
- (iv) Nothing else is a sentence of $\mathcal{L}$.

$\mathcal{L}_0$ is the *should*-free fragment of $\mathcal{L}$, so clause (2ii) here implies that *should* never occurs in the scope of another occurrence of *should*. This restriction may not be strictly necessary, but it does simplify the system considerably, and none of our data involve iterated *should*'s.

We adopt a ‘two-dimensional’ approach to presuppositions that separates the component of the theory that specifies presuppositions from the component that specifies semantic values. Nothing significant hangs on this choice; we adopt this approach because it is simple and clear and allows us to state enough of what we need to state about presuppositions. But it should not be difficult to implement our basic semantics for *should* using another approach to presuppositions.

Definition 2. For a sentence ϕ of $\mathcal{L}$, we define the *presupposition set of ϕ* , $\pi(\phi)$, as follows:

- 1. $\pi(p) = \emptyset$, for all atomic sentences p .
- 2. $\pi(\phi_\psi) = \pi(\phi) \cup \{\psi\}$
- 3. $\pi(\neg\phi) = \pi(\phi)$
- 4. $\pi(\phi \wedge \psi) = \pi(\phi) \cup \pi(\psi)$
- 5. $\pi(K\phi) = \pi(\phi)$
- 6. $\pi(\textit{should } \phi) = \pi(\phi)$

If Γ is a set of sentences, $\pi(\Gamma) = \{\phi : \phi \in \pi(\gamma) \text{ for some } \gamma \in \Gamma\}$.

Note that $\pi(\phi)$ and $\pi(\Gamma)$ are sets of *sentences*; we define corresponding sets of propositions below.

We present the specification of truth-conditions in two stages: first we present a semantics for the language $\mathcal{L}_0$ and then present the semantics for the full language $\mathcal{L}$. The semantics for $\mathcal{L}_0$ is a standard semantics for epistemic logic with the addition of the presupposition operator, which simply has the semantics of conjunction. The models for both languages are the same.

Definition 3. A *model* is a tuple $\mathcal{M} = (\mathcal{W}, \mathcal{K}, \mathcal{I})$, where $\mathcal{W}$ is a non-empty set of possible worlds, $\mathcal{K}$ is a reflexive binary relation on $\mathcal{W}$, and $\mathcal{I}$ is a function that maps an atomic sentence p of $\mathcal{L}_0/\mathcal{L}$ to a set of worlds $\mathcal{I}(p)$. A *pointed model* is a pair $(\mathcal{M}, w)$ consisting of a model and world from that model.

Definition 4. We define the semantic value of a sentence ϕ of $\mathcal{L}_0$ at a pointed model $(\mathcal{M}, w)$, as follows:

1. $\llbracket p \rrbracket^{\mathcal{M}, w} = 1$ iff $w \in \mathcal{I}(p)$
2. $\llbracket \neg\phi \rrbracket^{\mathcal{M}, w} = 1$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = 0$
3. $\llbracket (\phi \wedge \psi) \rrbracket^{\mathcal{M}, w} = 1$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = 1$ and $\llbracket \psi \rrbracket^{\mathcal{M}, w} = 1$
4. $\llbracket \phi\psi \rrbracket^{\mathcal{M}, w} = 1$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w} = 1$ and $\llbracket \psi \rrbracket^{\mathcal{M}, w} = 1$
5. $\llbracket K\phi \rrbracket^{\mathcal{M}, w} = 1$ iff: for all $w' \in \mathcal{K}(w)$, $\llbracket \phi \rrbracket^{\mathcal{M}, w'} = 1$

We define single-premise entailment for $\mathcal{L}_0$, and then define a notion of logical independence in terms of it.

Definition 5. If ϕ and ψ are $\mathcal{L}_0$ -sentences, then ϕ *0-entails* ψ , $\phi \models_0 \psi$, iff: for all pointed models $(\mathcal{M}, w)$, if $\llbracket \phi \rrbracket^{\mathcal{M}, w} = 1$, then $\llbracket \psi \rrbracket^{\mathcal{M}, w} = 1$.

Definition 6. If ϕ and ψ are $\mathcal{L}_0$ -sentences, then ϕ and ψ are *logically independent*, $\phi \perp\!\!\!\perp \psi$, iff neither 0-entails the other and neither 0-entails the negation of the other, i.e. the following all obtain:

- $\phi \not\models_0 \psi$
- $\phi \not\models_0 \neg\psi$
- $\psi \not\models_0 \phi$
- $\psi \not\models_0 \neg\phi$

Definition 7. Some notation:

1. Where ϕ is an $\mathcal{L}_0$ -sentence and $\mathcal{M}$ a model,

$$[\phi]^{\mathcal{M}} := \{w \in \mathcal{W} : \llbracket \phi \rrbracket^{\mathcal{M}, w} = 1\}.$$

2. Where ϕ is an $\mathcal{L}_0$ -sentence and $\mathcal{M}$ a model,

$$\Pi^{\mathcal{M}}(\phi) := \{[\psi]^{\mathcal{M}} \in \mathcal{P}(\mathcal{W}) : \psi \in \pi(\phi)\}.$$

3. Where ϕ is an $\mathcal{L}_0$ -sentence, Γ a set of $\mathcal{L}$ -sentences, and $\mathcal{M}$ a model,

$$\Pi_{\phi}^{\mathcal{M}}(\Gamma) := \{[\psi]^{\mathcal{M}} \in \mathcal{P}(\mathcal{W}) : \psi \in \mathcal{L}_0 \text{ \& } \psi \in \pi(\Gamma) \text{ \& } \psi \perp\!\!\!\perp \phi\}.$$

When the model is clear from context, we sometimes omit the superscripts in this notation (e.g. writing $[\phi]$ instead of $[\phi]^{\mathcal{M}}$). For a non-modal sentence ϕ , $[\phi]$ is the proposition expressed by ϕ , and $\Pi(\phi)$ is the set of ϕ 's presuppositions, construed as propositions rather than sentences. $\Pi_{\phi}(\Gamma)$ is the set of Γ 's non-modal presuppositions (construed as propositions) that are logically independent of ϕ .

We now present the semantics for the full language $\mathcal{L}$.

Definition 8. Let a *point* for $\mathcal{L}$ be a tuple $(\mathcal{M}, w, f, \succeq, \Gamma)$ consisting of a model $\mathcal{M}$, a world w in that model, a function f mapping each world to a non-empty set of worlds $f(w)$, a total preorder $\succeq$ on the set of worlds, and a set of $\mathcal{L}$ -sentences Γ . We assume that $\succeq$ verifies a ‘limit assumption’, so that for any non-empty set of worlds S :

$$\text{TOP}_{\succeq}(S) := \{w' \in S : \neg \exists w'' \in S \text{ s.t. } w'' \succ w'\} \neq \emptyset.$$

Definition 9. We define the semantic value of a sentence ϕ of $\mathcal{L}$ at a point $(\mathcal{M}, w, f, \succeq, \Gamma)$, $\llbracket \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma}$, as follows:

1. $\llbracket p \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ iff $w \in \mathcal{I}(p)$
2. $\llbracket \neg \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 0$
3. $\llbracket (\phi \wedge \psi) \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ and $\llbracket \psi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$
4. $\llbracket \phi \psi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ iff $\llbracket \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ and $\llbracket \psi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$
5. $\llbracket K\phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ iff: for all $w' \in \mathcal{K}(w)$, $\llbracket \phi \rrbracket^{\mathcal{M}, w', f, \succeq, \Gamma} = 1$
6. $\llbracket \text{should } \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$ iff for all $w' \in \text{TOP}_{\succeq}(D(\phi))$, $\llbracket \phi \rrbracket^{\mathcal{M}, w', f, \succeq, \Gamma} = 1$, where

$$D(\phi) = f(w) \cap \bigcap \Pi^{\mathcal{M}}(\phi) \cap \bigcap \Pi_{\phi}^{\mathcal{M}}(\Gamma)$$

As in the text, we assume that if S is a set of propositions and S is empty, then $\bigcap S = \mathcal{W}$.

Finally, we define entailment and consistency for $\mathcal{L}$:

Definition 10. If Γ is a set of $\mathcal{L}$ -sentences and ϕ is an $\mathcal{L}$ -sentence, then Γ *entails* ϕ , $\Gamma \models \phi$, iff: for all points $(\mathcal{M}, w, f, \succeq, \Delta)$ such that $\Delta = \Gamma$, if for all $\gamma \in \Gamma$, $\llbracket \gamma \rrbracket^{\mathcal{M}, w, f, \succeq, \Delta} = 1$, then $\llbracket \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Delta} = 1$.

Note that we only consider points whose discourse parameter Δ is set equal to the premise set Γ . When the premise set Γ is a singleton $\{\psi\}$, we write $\psi \models \phi$ instead of $\{\psi\} \models \phi$.

Definition 11. If Γ is a set of $\mathcal{L}$ -sentences, then Γ is *consistent* iff there is a point $(\mathcal{M}, w, f, \succeq, \Delta)$ such that $\Delta = \Gamma$ and for all $\gamma \in \Gamma$, $\llbracket \gamma \rrbracket^{\mathcal{M}, w, f, \succeq, \Delta} = 1$. An $\mathcal{L}$ -sentence ϕ is consistent iff $\{\phi\}$ is consistent.

A few lemmas will be useful in what follows:

Lemma 1. If $\phi \in \mathcal{L}_0$, then for any point $(\mathcal{M}, w, f, \succeq, \Gamma)$,

$$\llbracket \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = \llbracket \phi \rrbracket^{\mathcal{M}, w}.$$

Proof. By induction on the complexity of $\mathcal{L}_0$ -sentences. □

Lemma 2. If $\phi \in \mathcal{L}_0$, then $\pi(\phi) \subseteq \mathcal{L}_0$.

Proof. By induction on the complexity of $\mathcal{L}_0$ -sentences. □

Lemma 3. For any formula ϕ of $\mathcal{L}_0$ and any point $(\mathcal{M}, w, f, \succeq, \Gamma)$ with $\Gamma = \{\text{should } \phi\}$,

$$D(\phi) = f(w) \cap \bigcap \Pi(\phi).$$

Proof. Let $\phi \in \mathcal{L}_0$ and let $(\mathcal{M}, w, f, \succeq, \Gamma)$ be a point with $\Gamma = \{\text{should } \phi\}$. Then

$$D(\phi) = f(w) \cap \bigcap \Pi(\phi) \cap \bigcap \Pi_\phi(\{\text{should } \phi\}).$$

So it will suffice to show:

$$\bigcap \Pi(\phi) \cap \bigcap \Pi_\phi(\{\text{should } \phi\}) = \bigcap \Pi(\phi).$$

Recall that:

$$\Pi(\phi) = \{[\gamma] : \gamma \in \pi(\phi)\}.$$

And:

$$\Pi_\phi(\{\text{should } \phi\}) = \{[\gamma] : \gamma \in \mathcal{L}_0 \text{ \& } \gamma \in \pi(\{\text{should } \phi\}) \text{ \& } \gamma \perp\!\!\!\perp \phi\}.$$

Since $\pi(\{\text{should } \phi\}) = \pi(\phi)$, we have:

$$\Pi_\phi(\{\text{should } \phi\}) = \{[\gamma] : \gamma \in \mathcal{L}_0 \text{ \& } \gamma \in \pi(\phi) \text{ \& } \gamma \perp\!\!\!\perp \phi\}.$$

It follows that:

$$\Pi_\phi(\{\text{should } \phi\}) \subseteq \Pi(\phi).$$

Now if A and B are sets of sets and $A \subseteq B$, then $\bigcap B \subseteq \bigcap A$. And in that case $\bigcap B \cap \bigcap A = \bigcap B$.

So it follows that:

$$\bigcap \Pi(\phi) \cap \bigcap \Pi_\phi(\{\text{should } \phi\}) = \bigcap \Pi(\phi),$$

which is what we needed to show. □

We can now state some facts about the inferences of interest.

Fact 1 (Known Violation Invalidity). $\text{should } Kb_b \not\models \text{should } b$.

The proof of this is omitted here as it can be reconstructed from the argument given in Section 5.3.

Proposition 1 (Restricted Inheritance). Suppose ϕ and ψ are $\mathcal{L}_0$ -sentences such that $\pi(\phi) = \pi(\psi)$. Then if $\phi \models_0 \psi$, $\text{should } \phi \models \text{should } \psi$.

Proof. Suppose ϕ and ψ are as described above and that $\phi \models_0 \psi$. Let $(\mathcal{M}, w, f, \succeq, \Gamma)$ be a point with $\Gamma = \{\text{should } \phi\}$. Suppose $\llbracket \text{should } \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$. We need to show that $\llbracket \text{should } \psi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$.

Since $\llbracket \text{should } \phi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$, every world $w' \in \text{TOP}_{\succeq}(D(\phi))$ is such that $\llbracket \phi \rrbracket^{\mathcal{M}, w', f, \succeq, \Gamma} = 1$, where

$$D(\phi) = f(w) \cap \bigcap \Pi(\phi) \cap \bigcap \Pi_{\phi}(\{\text{should } \phi\}).$$

By **Lemma 3**:

$$D(\phi) = f(w) \cap \bigcap \Pi(\phi).$$

To show $\llbracket \text{should } \psi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$, we need to show that every world $w' \in \text{TOP}_{\succeq}(D(\psi))$ is such that $\llbracket \psi \rrbracket^{\mathcal{M}, w', f, \succeq, \Gamma} = 1$, where

$$D(\psi) = f(w) \cap \bigcap \Pi(\psi) \cap \bigcap \Pi_{\psi}(\{\text{should } \phi\}).$$

Now

$$\Pi_{\psi}(\{\text{should } \phi\}) = \{[\gamma] : \gamma \in \mathcal{L}_0 \text{ \& } \gamma \in \pi(\{\text{should } \phi\}) \text{ \& } \gamma \perp\!\!\!\perp \psi\}.$$

Since $\pi(\text{should } \phi) = \pi(\phi) = \pi(\psi)$, it follows that:

$$\Pi_{\psi}(\{\text{should } \phi\}) = \{[\gamma] : \gamma \in \mathcal{L}_0 \text{ \& } \gamma \in \pi(\psi) \text{ \& } \gamma \perp\!\!\!\perp \psi\}.$$

Thus,

$$\Pi_{\psi}(\{\text{should } \phi\}) \subseteq \Pi(\psi).$$

Using the same set-theoretic facts we used in the proof of **Lemma 3**, it follows that:

$$\bigcap \Pi(\psi) \cap \bigcap \Pi_{\psi}(\{\text{should } \phi\}) = \bigcap \Pi(\psi).$$

So

$$D(\psi) = f(w) \cap \bigcap \Pi(\psi).$$

Since $\pi(\psi) = \pi(\phi)$, $\Pi(\psi) = \Pi(\phi)$. So we have:

$$D(\psi) = f(w) \cap \bigcap \Pi(\phi) = D(\phi).$$

So let w' be an element of $\text{TOP}_{\succeq}(D(\psi))$. Then $w' \in \text{TOP}_{\succeq}(D(\phi))$, and so $\llbracket \phi \rrbracket^{\mathcal{M}, w', f, \succeq, \Gamma} = 1$. Since $\phi \in \mathcal{L}_0$, it follows from **Lemma 1** that $\llbracket \phi \rrbracket^{\mathcal{M}, w'} = 1$. Since $\phi \models_0 \psi$, it follows that $\llbracket \psi \rrbracket^{\mathcal{M}, w'} = 1$.

So by **Lemma 1** again, it follows that $\llbracket \psi \rrbracket^{\mathcal{M}, w', f, \succeq, \Gamma} = 1$. Since w' was an arbitrary element of $\text{TOP}_{\succeq}(D(\psi))$, it follows that $\llbracket \text{should } \psi \rrbracket^{\mathcal{M}, w, f, \succeq, \Gamma} = 1$, which is what we needed to show. $\square$

Corollary 1 (Procrastinate Validity). $\text{should } (i \wedge r) \models \text{should } i$

Proof. By **Proposition 1**, since $\{(i \wedge r), i\} \subseteq \mathcal{L}_0$, $(i \wedge r) \models_0 i$, and $\pi(i \wedge r) = \emptyset = \pi(i)$. $\square$

Fact 2. $\text{should } (Kb_b \wedge a) \models \text{should } a$

The proof of this is omitted here, but can be reconstructed from the argument given in Section 5.3.

Fact 3. $\text{should } (Kb_b \wedge a) \not\models \text{should } (b \wedge a)$

The proof of this is similar to the argument for **Fact 1** and is left to the reader.

Finally, the proof of the following two claims can be reconstructed from the discussion in Section 5.4:

Fact 4. $(\text{should } Kb_b \wedge \text{should } \neg b)$ is consistent.

Corollary 2 (Agglomeration Failure). $\{\text{should } Kb_b, \text{should } \neg b\} \not\models \text{should } (Kb_b \wedge \neg b)$.

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